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DEPARTMENT OF BIOLOGICAL AND PHYSICAL SCIENCES

ASYNCHRONOUS ACTIVITY #2

STAT 2A (Applied Statistics)

GENERAL DIRECTIONS: Read and answer each item as directed. Write your answers on an A4-sized bond paper and show your complete computation for each item. All computations and answers must be **HANDWRITTEN**.

1. A bag contains 5 red, 4 blue, and 3 green marbles. If three marbles will be drawn in succession, what is the probability that all marbles are red if:
    a. they are drawn with replacement; and
    b. they are drawn without replacement?

2. A standard PIN number is made up of 4 digits. What is the probability that a randomly generated PIN number starts with either 4 or 6 and ends with an even number?

3. For a school activity, a committee of six shall be formed out of students from various sections. If there are 5 student-candidates from section A, 6 from section B, and 4 from section C, what is the probability that the committee formed includes exactly two students from each section?

4. Towards the end of a basketball match, these are the current shooting performances of the players of CvSU-Imus:

Player Jersey #	Field Goals	3-points	Free Throws
16	8 / 15	3 / 7	4 / 5
28	5 / 10	1 / 4	2 / 2
67	6 / 11	0 / 2	3 / 4
77	3 / 8	2 / 5	1 / 2
96	2 / 4	1 / 1	1 / 1

Note: 3-point statistics is a subset of field goals statistics

A. What is the probability that any player from the team will successfully pull off a three-point shot in the current rally?

B. During the last 10 seconds of the game, the teams have their scores tied and the ball is in the possession of CvSU-Imus. Excluding all other factors aside from shooting performances, who has the highest probability of scoring a shot to make the team win?

C. Due to some in-game reasons, an intentional foul was made towards player # 16 and was given two free throw shots. Suppose the first shot failed, what is the probability that the second shot will be successful?